

1 Notation

We model four events each year: growth, colonization, fizzle and burnout.

Let (k, t) be the time index, where k represent the year index and t represent the season index, where $t = 0$ for the time at the beginning of the year, $t = 1$ for the time after growth, $t = 2$ for the time after colonization, $t = 3$ for the time after fizzle and $t = 4$ for the time after burnout. So $(k, 0)$ represents the same time as $(k - 1, 4)$

Assume there are n patches and we label them with p_i . For each patch we use $(\mathbb{1}_i^{(k,t)}, N_i^{(k,t)})$ to describe its state. $N_i^{(k,t)}$ represents population size in patch p_i at time point (k, t) , and $\mathbb{1}_i^{(k,t)}$ is defined as:

$$\mathbb{1}_i^{(k,t)} = \begin{cases} 1, & \text{if } p_i \text{ is infected at time } (k, t), \\ 0, & \text{otherwise} \end{cases}$$

Let $U \sim \text{Uniform}(0, 1)$ be a random variable uniformly distributed over $(0, 1)$, and we have

$$\mathbb{1}_{U \leq P} = \begin{cases} 1, & \text{if } U \leq P, \\ 0, & \text{otherwise.} \end{cases}$$

2 Growth

We can assume that reproduction of rats follows logistic growth, i.e.,

$$N_i^{(k,1)} = N_i^{(k,0)} + rN_i^{(k,0)}\left(1 - \frac{N_i^{(k,0)}}{K}\right)$$

where r is the growth rate and K is the carry capacity of a patch. We currently assume that all patches are of equal area and have the same carrying capacity.

3 Colonization

3.1 Population size

We assume that each individual rat move with probability c and land in a random patch. We ignore different distances and assume that all patches are of equal availability to a moving rat. So

$$N_i^{(k,2)} = (1 - c)N_i^{(k,1)} + \frac{c \sum_i N_i^{(k,1)}}{n}$$

The first term is population staying in the patch and the second term is population moving into the patch.

3.2 Infection

3.2.1 version 1 (based on death number)

When an infected host dies the fleas are released and can travel between patches (such as being carried by a moving individual) and infect hosts in other patches.

Let z be the final size. The number of individuals that died of the disease in the previous year in a patch is $zDN_i^{(k-1,3)}$ if patch p_i is infected, and 0 if not. So the total number of death is $\sum_i zDN_i^{(k-1,3)}\mathbb{1}_i^{(k-1,3)}$.

Assume the probability of a susceptible patch getting infected follows hazard model

$$P_I^{(k)} = 1 - e^{-\frac{\alpha z D \sum_i N_i^{(k-1,3)} \mathbb{1}_i^{(k-1,3)}}{nK}}$$

where α is a scaling constant.

3.2.2 version 2 (based on colonization)

Let z be the final size. The number of individuals that was infected in the previous year in a patch is $zN_i^{(k-1,3)}$ if patch p_i is infected, and 0 if not, and they move with probability c . So the total number of infected rats moving is

$$cz \sum_i N_i^{(k-1,3)} \mathbb{1}_i^{(k-1,3)}$$

and the expected number of infected rats moving into a patch is

$$\frac{cz \sum_i N_i^{(k-1,3)} \mathbb{1}_i^{(k-1,3)}}{n}$$

Assume the probability of a susceptible patch getting infected follows hazard model

$$P_I^{(k)} = 1 - e^{-\frac{\alpha cz \sum_i N_i^{(k-1,3)} \mathbb{1}_i^{(k-1,3)}}{n}}$$

where α is the probability of a susceptible patch getting infected caused by an infected individual moving in.

And

$$\mathbb{1}_i^{(k,2)} = 1 - (1 - \mathbb{1}_i^{(k,1)})\mathbb{1}_{U < P_I^{(k)}}$$

4 Fizzle

Fizzle probability is

$$P_f = \frac{1}{\mathcal{R}_0}$$

And

$$\mathbb{1}_i^{(k,3)} = \mathbb{1}_i^{(k,2)}\mathbb{1}_{U < P_f}$$

Population size does not change.

5 Burnout

In infected patches, the population should be reduced:

$$N_i^{(k,4)} = (1 - zD\mathbb{1}_i^{(k,3)})N_i^{(k,3)}$$

Burnout probability if a patch didn't fizzle is

$$P_b = q^{Ny^*}$$

(q and y^* are defined in burnout paper)

And

$$\mathbb{1}_i^{(k,4)} = \mathbb{1}_i^{(k,3)}\mathbb{1}_{U < P_b}$$